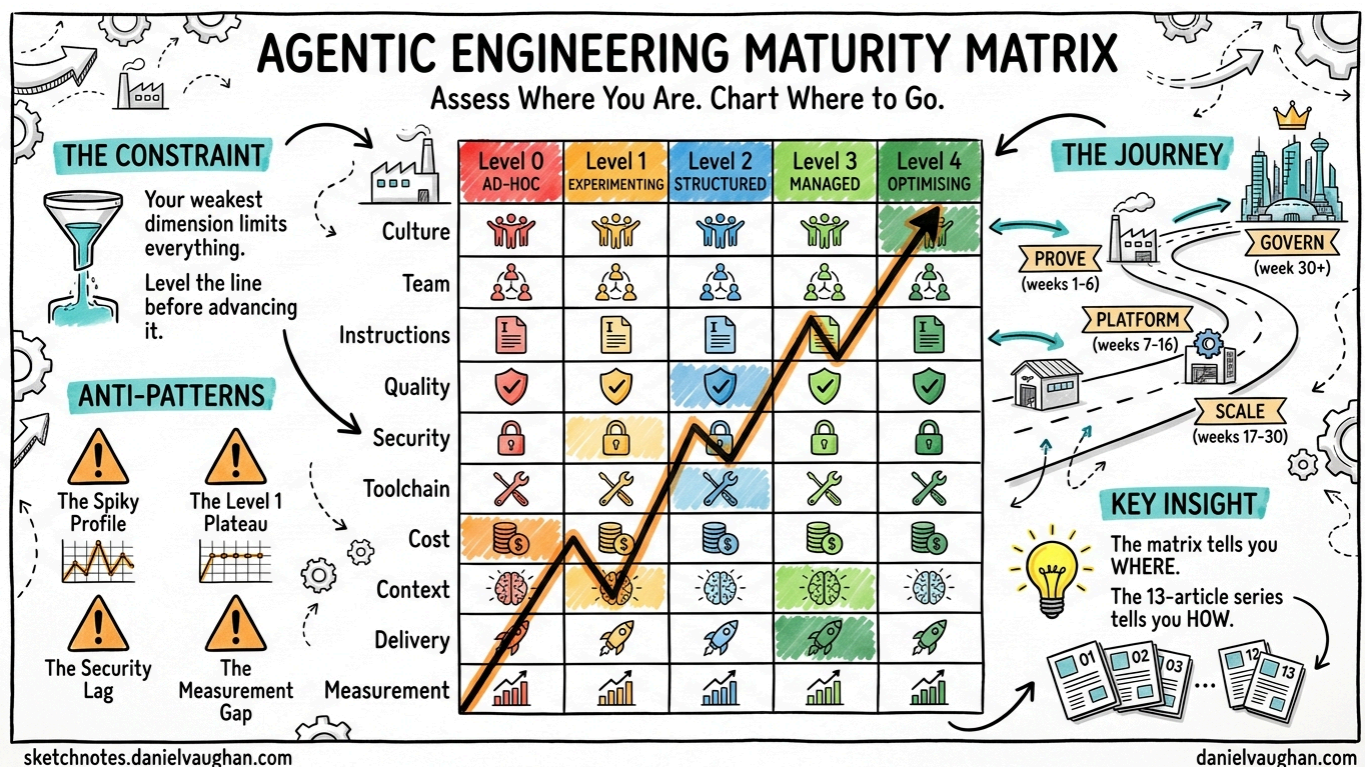


The Agentic Engineering Maturity Matrix: Assessing Your Organisation's AI Readiness



The Agentic Engineering Series. From experiment to enterprise. This is article 13 of 13. *This article gives you the assessment frame: a maturity matrix that shows where your organisation stands now and what has to improve next.* Previous: The Scaling Playbook | Series overview

Series context: This is article 13 of 13 in *From Experiment to Factory*. The series has covered the thesis, platform, blueprint, quality gate, team model, toolchain, risk, engine, guardrails, efficiency, business case and rollout. This final piece is the assessment layer: how to score what is actually in place, find the bottleneck and choose the next move.

Container Solutions changed cloud consulting with a single grid: the Cloud Native Maturity Matrix.¹ It gave clients a shared way to answer two practical questions. Where are we now? What has to improve next?

Agentic engineering needs the same thing. Most organisations can list the tools they have tried. Far fewer can say whether those tools sit inside a repeatable system or inside a growing pile of cognitive debt.

That is the short answer. The Agentic Engineering Maturity Matrix gives you a common language for scoring how disciplined your adoption really is. It measures 10 dimensions across five levels and forces you to look for the weakest point, because that point limits everything else.

If you remember one rule from this article, remember this one: do not score intention. Score observable evidence. A team is not 'managed' because it plans to formalise `AGENTS.md` next quarter. It is managed when shared templates, governance and upkeep already exist.

Quick self-assessment

Before you work through the full matrix, use these checks to find your likely floor. Count the yes answers at each level. The highest level where every answer is yes is your approximate current position.

Level 1, experimenting

- Do fewer than three developers use AI coding tools regularly?

- Is there no `AGENTS.md` file in any repository?
- Is there no token or cost tracking for AI tool usage?

If the answer is yes to all three, you are at Level 0 or early Level 1.

Level 2, structured

- Does at least one team have a versioned `AGENTS.md` with boundary rules such as Always, Ask First and Never?
- Are tests written before agent execution on critical paths?
- Is there a defined pod or team structure for agentic work, rather than isolated individuals using tools?

If the answer is yes to all three, you have reached Level 2.

Level 3, managed

- Do multiple teams share a common `AGENTS.md` template library or platform layer?
- Are per-team token budgets in place with escalation triggers?
- Is agent output subject to automated security scanning through hooks, SAST or equivalent controls?

If the answer is yes to all three, you have reached Level 3.

Level 4, optimising

- Is cost per merged PR tracked and trending down?
- Does a knowledge codification cycle feed lessons back into `AGENTS.md` and memory systems?
- Are A/B experiments run on methodology changes, with ROI reported at board level?

If the answer is yes to all three, you are at Level 4.

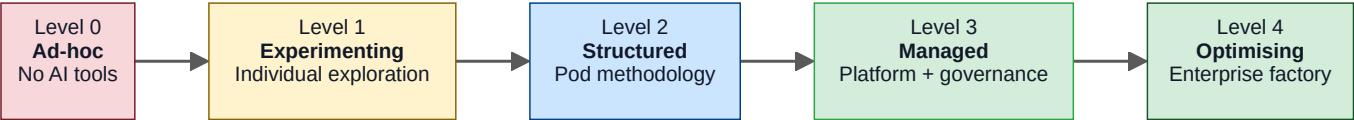
The quick check is only a starting point. The full matrix matters because organisations rarely mature evenly.

The matrix

The matrix scores organisations across 10 dimensions and five maturity levels. Each cell should describe something an assessor can verify. It is not a wish list.

THE FIVE LEVELS

Level	Name	Description
0	Ad-hoc	No systematic approach. AI tools are absent or used opportunistically.
1	Experimenting	Individual developers explore AI tools. Team standards do not exist yet.
2	Structured	One or more teams use a defined agentic engineering method.
3	Managed	Multiple teams share platform, governance and measurement.
4	Optimising	The organisation runs an enterprise factory with continuous improvement and ROI tracking.



The dimensions below come from the whole series. Each one maps to an article that explains how to move it forward.

The complete matrix

1. ENGINEERING CULTURE

How does the organisation think about AI-assisted development?

Level	State	Observable evidence
0 — Ad-hoc	No AI awareness	Developers are unaware of or openly resistant to AI tools
1 — Experimenting	Curiosity-driven	Individuals try tools, there is no policy and vibe coding is common
2 — Structured	Method-led	A team adopts compound engineering, planning and review matter more than generation speed
3 — Managed	Engineering discipline	The organisation recognises that code generation is cheap and verification, design and context are the bottleneck
4 — Optimising	Continuous learning	Knowledge codification is standard, lessons feed back into the system and skill matrices are maintained

Key series reference: Article 02 — The Thesis

2. TEAM STRUCTURE

How are humans and agents organised?

Level	State	Observable evidence
0 — Ad-hoc	Traditional teams	No role differentiation for AI work, standard dev/QA/PM structure
1 — Experimenting	Solo plus agent	Individual developers use agents as personal assistants
2 — Structured	Agentic pod	Three-person pods, Context Architect, Value Engineer and Quality Engineer, each using agents and each accountable for output
3 — Managed	Multiple pods	Parallel pods share a platform layer, Context Architects share knowledge across pods and there are no passive coordinators
4 — Optimising	Pod ecosystem	Pods self-organise around work, a platform team maintains shared infrastructure and scale comes from launching pods rather than growing them

Key series references: Article 03 — The Team, Article 12 — The Adoption

3. AGENT INSTRUCTIONS

How are agents governed and directed?

Level	State	Observable evidence
0 — Ad-hoc	No instructions	Developers prompt agents ad hoc, with no persistent configuration
1 — Experimenting	Personal prompts	Developers keep private prompt templates, nothing is versioned
2 — Structured	<code>AGENTS.md</code>	Each project has a versioned boundary file defining constraints, forbidden actions and workflow rules
3 — Managed	Template library	Teams share <code>AGENTS.md</code> templates, organisational golden paths and an ETH Zurich-aligned structure
4 — Optimising	Living documents	<code>AGENTS.md</code> files are updated through the knowledge codification cycle and agent behaviour improves from project to project

Key series reference: Article 05 — The Blueprint

4. QUALITY AND TESTING

How is agent-generated code verified?

Level	State	Observable evidence
0 — Ad-hoc	Manual testing	Tests are written afterwards, if at all, and there are no AI-specific quality gates
1 — Experimenting	Basic tests	Some unit tests exist, agents sometimes write tests and verification remains inconsistent
2 — Structured	TDAD	Test-Driven Agent Development is in use, tests are written before agent execution and agents must prove correctness
3 — Managed	Quality gates	Pre-commit hooks enforce passing tests, parallel review agents cover security, performance and correctness and the Quality Engineer can veto structurally weak output
4 — Optimising	Continuous verification	Edge-case generation, mutation testing and escape-rate tracking are standard and trend data improves quarter by quarter

Key series references: Article 04 — The Quality Gate, Article 10 — The Warning

5. SECURITY AND GUARDRAILS

How is AI-generated code secured structurally?

Level	State	Observable evidence
0 — Ad-hoc	No AI security policy	Agent output is treated exactly like human output, with no extra controls
1 — Experimenting	Awareness	Teams know about sandbox modes, some developers use <code>suggest</code> mode and enforcement is absent
2 — Structured	Structural enforcement	Sandbox controls, hooks and <code>requirements.toml</code> boundaries are in place and the Quality Engineer runs untrusted mode where appropriate
3 — Managed	Enterprise guardrails	MDM-deployed configs, network controls, pre-commit security scanning, audit logs and compliance documentation are present
4 — Optimising	Continuous protection	Vulnerability scanning, real-time detection, policy-as-code and regular red-team exercises cover agent workflows

Key series reference: Article 07 — The Guardrails

6. TOOLCHAIN AND PLATFORM

What tools are in use and how are they managed?

Level	State	Observable evidence
0 — Ad-hoc	No AI tools	Or individual subscriptions with no standardisation
1 — Experimenting	Single-tool trial	One tool, such as Copilot or Cursor, is used informally and there is no evaluation frame
2 — Structured	Evaluated selection	A primary tool is chosen through structured evaluation, a backup exists and a local model option has been tested
3 — Managed	Multi-tool strategy	A primary CLI agent, local models and IDE integration coexist, model routing is configured and no single vendor is a hard dependency
4 — Optimising	AI gateway	A central gateway routes agent traffic, supports provider switching, exposes telemetry and enforces policy at infrastructure level

Key series references: Article 09 — The Toolchain, Article 06 — The Engine, Article 11 — The Business Case

7. COST GOVERNANCE

How are AI costs tracked, managed and optimised?

Level	State	Observable evidence
0 — Ad-hoc	No tracking	AI costs are buried in general software spend and nobody has per-developer visibility
1 — Experimenting	Subscription-based	Flat-rate plans dominate, task-level cost awareness is absent and subsidy assumptions remain untested
2 — Structured	Token awareness	API billing is active, hooks log per-session cost, developers can see their own consumption and model routing reduces cost
3 — Managed	Budget governance	Per-team budgets, escalation triggers, weekly telemetry, access tiers and quarterly efficiency reviews are in place
4 — Optimising	ROI-driven	The organisation measures cost, efficiency, productivity and business outcomes together, tracks cost per merged PR and balances hardware and cloud spend deliberately

Key series reference: Article 11 — The Business Case

8. CONTEXT MANAGEMENT

How is the agent's working memory managed across sessions and teams?

Level	State	Observable evidence
0 — Ad-hoc	No management	Sessions run until context exhausts, <code>.codexignore</code> is absent and compaction regressions are common
1 — Experimenting	Basic hygiene	<code>.codexignore</code> exists, developers know context limits and manual compaction happens sometimes
2 — Structured	Active management	Teams use subagent delegation, early compaction, role-tuned reasoning effort and structured memory extraction
3 — Managed	Organisational memory	Codebase memory systems, knowledge graphs, cross-session persistence and documented context architecture are in place
4 — Optimising	Compounding knowledge	Every cycle codifies lessons, memory systems measurably improve outcomes and context cost is treated as part of token economics

Key series reference: Article 08 — The Operations

9. DELIVERY AND PROCESS

How does the engineering process incorporate AI agents?

Level	State	Observable evidence
0 — Ad-hoc	Traditional process	Standard Agile or Scrum continues with no AI-specific integration and no distinction in review practice
1 — Experimenting	Agent-augmented	Agents generate code, the surrounding process barely changes and PR scrutiny often weakens because 'the AI wrote it'
2 — Structured	Compound engineering	Teams plan before implementation, review as rigorously as they would review junior output and codify what they learn
3 — Managed	Automated gates	CI/CD enforces AI-specific checks, pre-merge hooks are active, review agents run in parallel and cadence improves without sacrificing quality
4 — Optimising	Continuous delivery	Lead time, defect escape rate and delivery capacity are tracked and used to refine the process continuously

Key series references: Article 02 — The Thesis, Article 01 — The Platform

10. MEASUREMENT AND LEARNING

How does the organisation measure impact and learn from the data?

Level	State	Observable evidence
0 — Ad-hoc	No measurement	'It feels faster' is the only signal, there are no baselines and no before-and-after comparison
1 — Experimenting	Anecdotal	Satisfaction surveys and informal speed comparisons exist, but the productivity paradox remains invisible
2 — Structured	Baseline plus delta	Cycle time, defect rate and PR turnaround are measured before and after adoption, with speed paired to quality
3 — Managed	Four-layer metrics	Cost, efficiency, productivity and business outcomes are connected and reviewed quarterly, with the METR paradox considered in the design
4 — Optimising	Data-driven improvement	Telemetry drives process changes, methodology experiments run regularly and ROI is reported at board level

Key series references: Article 11 — The Business Case, Article 12 — The Adoption

How to assess an organisation

The matrix works because it is simple to run and hard to fake.

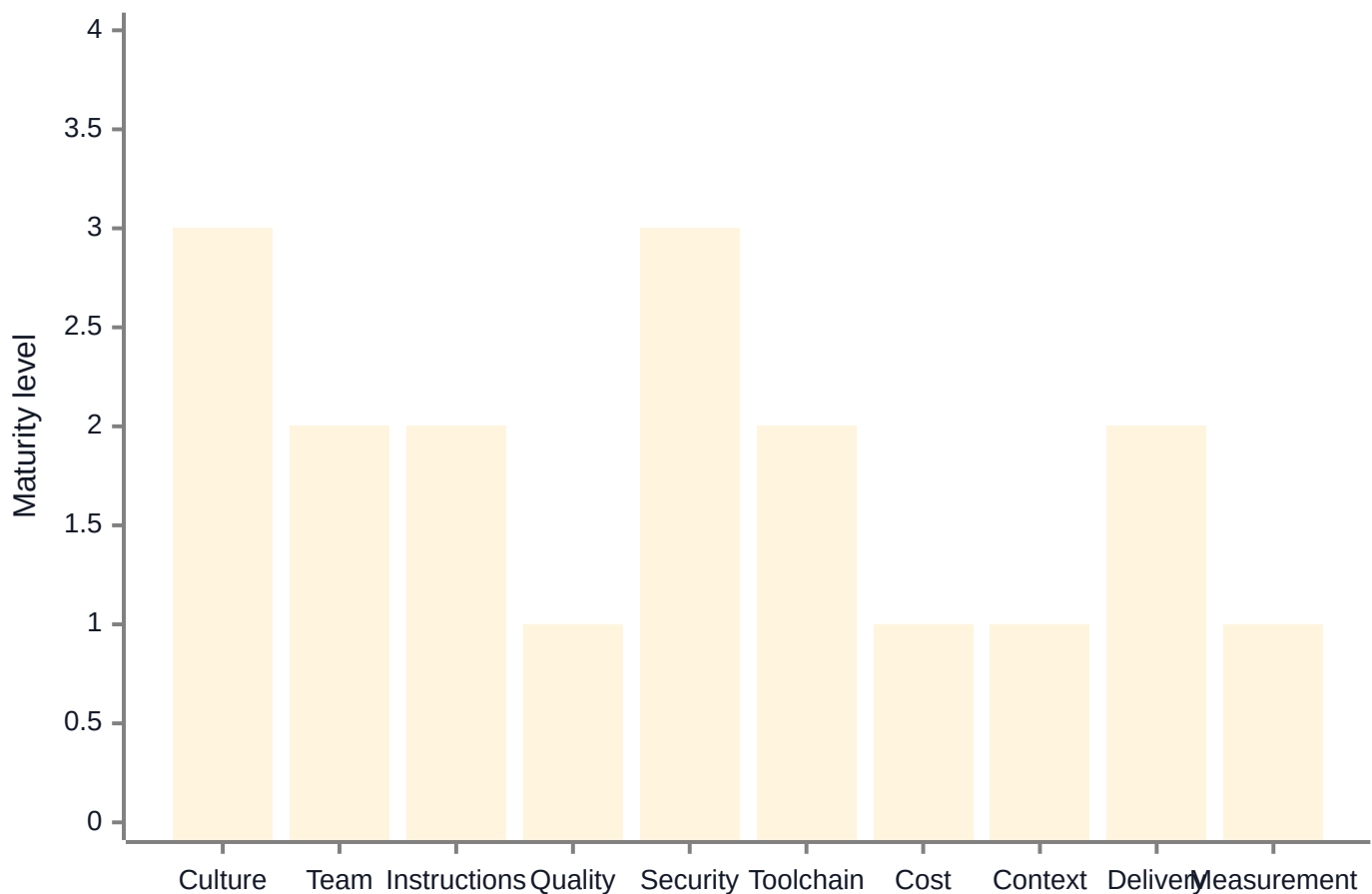
STEP 1: SCORE EACH DIMENSION

For each of the 10 dimensions, choose the level that best matches the current observable state. Score what exists now, not what is planned. If an AGENTS.md file exists but nobody maintains it, that remains Level 2 rather than Level 3.

STEP 2: DRAW THE LINE

Plot the scores and look at the shape. This is the Container Solutions insight applied directly: do not average the result away, look at the unevenness.¹

Example: enterprise at mixed maturity



STEP 3: IDENTIFY THE CONSTRAINT

The Theory of Constraints applies cleanly here.² Your overall maturity is capped by the weakest dimension. In the example above, Quality, Cost, Context and Measurement all sit at Level 1. It does not matter that Culture is at Level 3 if the organisation still cannot measure whether the tools are helping.

This is why the goal is not to maximise everything at once. The goal is to level the line. Raise the weakest dimensions first, because that is where the leverage is.

STEP 4: MAP THE PATH

Once you know the weak points, you can map them to the practices in this series.

Move	What you need	Primary reference
1 → 2	Compound engineering, <code>AGENTS.md</code> , a first pod and TDAD basics	Articles 01, 03, 04, 05
2 → 3	A shared platform layer, enterprise guardrails, multiple pods, token governance and four-layer metrics	Articles 09, 11, 12
3 → 4	An AI gateway, ROI-led optimisation, continuous learning and knowledge compounding	Articles 10, 11, 12

A consulting assessment protocol

If you are using the matrix as a consulting or leadership tool, run it as a short evidence-led engagement rather than a survey.

PRE-ASSESSMENT, ONE DAY

1. Interview the core stakeholders: CTO, VP engineering, two or three team leads and two or three developers.
2. Collect the artefacts: `AGENTS.md` files, hook configurations, CI/CD pipelines, cost reports and team-structure documents.

3. Request the baseline metrics: cycle time, defect rate, PR turnaround and AI-tool adoption data.

ASSESSMENT WORKSHOP, HALF A DAY

Run a facilitated workshop with six to 10 people from leadership, team leads and frontline developers.

For each dimension:

1. Read the five level descriptions aloud.
2. Ask participants to score independently.
3. Discuss the gaps in scoring, because those disagreements reveal the real state fastest.
4. Agree a consensus score and record the evidence behind it.

Current state
Score all 10 dimensions



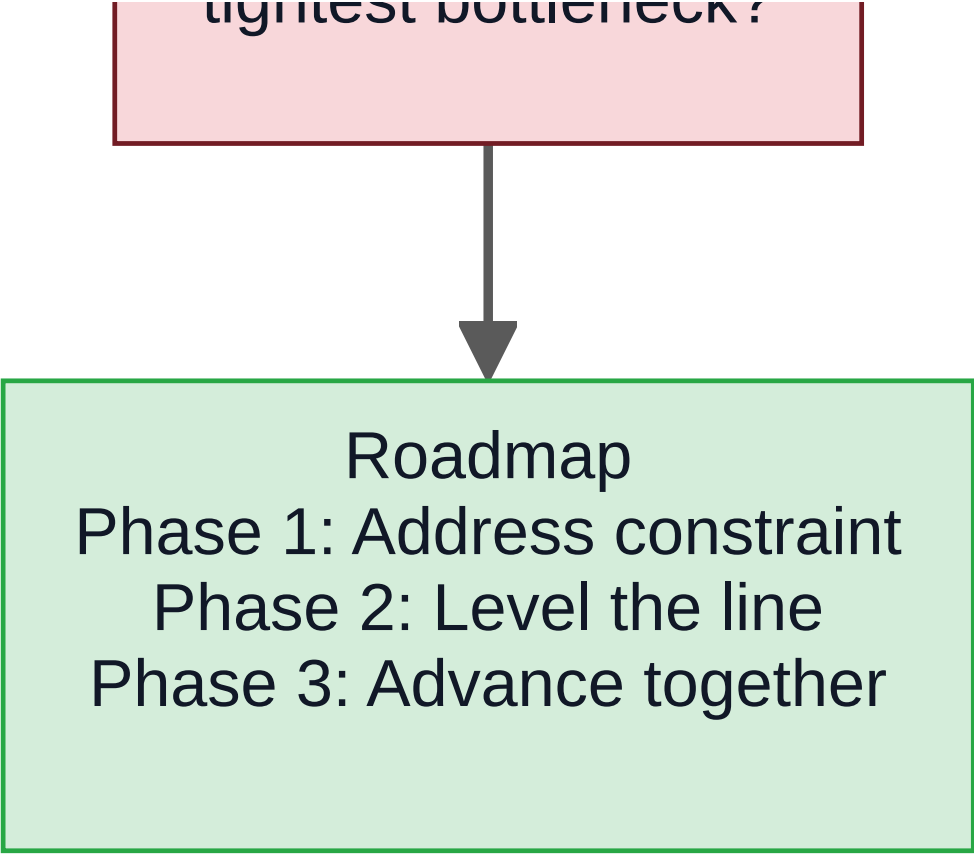
Target state
Where do you need to be
in 6, 12 or 18 months?



Gap identification
Which dimensions are
furthest from target?



Constraint analysis
Which gap is the
tightest bottleneck?



ROADMAP GENERATION

Use the four adoption phases from Article 12 — The Scaling Playbook to turn the scores into a plan.

Current move	Target phase	Timeline	Key actions
0 → 1	PROVE prep	2–4 weeks	Select tools, train the pilot group and identify the first team
1 → 2	PROVE	4–6 weeks	Stand up the first pod, adopt compound engineering, write <code>AGENTS.md</code> and establish baselines
2 → 3	PLATFORM + SCALE	10–20 weeks	Build the shared layer, launch more pods, deploy governance and run the 11-week training cycle
3 → 4	GOVERN + OPTIMISE	20+ weeks	Add the AI gateway, formal ROI measurement and a continuous improvement loop

What good looks like

Not every organisation needs Level 4 everywhere. The right target depends on scale and risk.

Organisation type	Recommended target	Rationale
Startup, under 50 engineers	Level 2 across all dimensions	One pod, a structured method and basic governance are enough
Scale-up, 50–200 engineers	Level 3, with a few dimensions at Level 4	Multiple pods, a platform layer and managed governance become necessary
Enterprise, 200+ engineers	Level 3–4 across all dimensions	At this point you need a full factory, an AI gateway and ROI-led management
Regulated industry	Security and Quality at Level 4 minimum	Compliance pressure makes those dimensions non-negotiable

ANTI-PATTERNS TO WATCH

The spiky profile

One dimension sits at Level 4 and several others sit at Level 1. Usually one strong internal champion has pushed a single area forward without broader support. The result looks impressive in a slide deck and brittle in practice.

The Level 1 plateau

Everybody is experimenting and nobody is structuring. This is one of the most common profiles in early 2026. Teams feel faster, but because they are not measuring properly they do not see the productivity paradox.³

The security lag

Culture and Delivery reach Level 3 while Security stays at Level 1. This is the Amazon pattern.⁴ Sophisticated use arrives before governance, and the production incident becomes the forcing function.

The measurement gap

Everything else advances while Measurement stays at Level 0 or 1. Without measurement you cannot prove ROI. Without ROI you cannot defend budget. That is how programmes die when cost pressure arrives.⁵

Put the matrix into your regular review cycle

The matrix is not a one-off workshop. It belongs inside the engineering cadence.

QUARTERLY MATURITY REVIEW

Activity	Frequency	Participants	Output
Full matrix reassessment	Quarterly	Engineering leadership and pod leads	Updated scores and trend analysis
Constraint review	Quarterly	The same group plus relevant dimension owners	Updated roadmap and resource decisions
Dimension deep-dives	Monthly, rotating	Dimension owner and assessment team	Specific improvement actions
Score tracking	Continuous	Automated where possible	A dashboard showing dimension trends

TRACKING TEMPLATE

Maintain a simple working document like this:

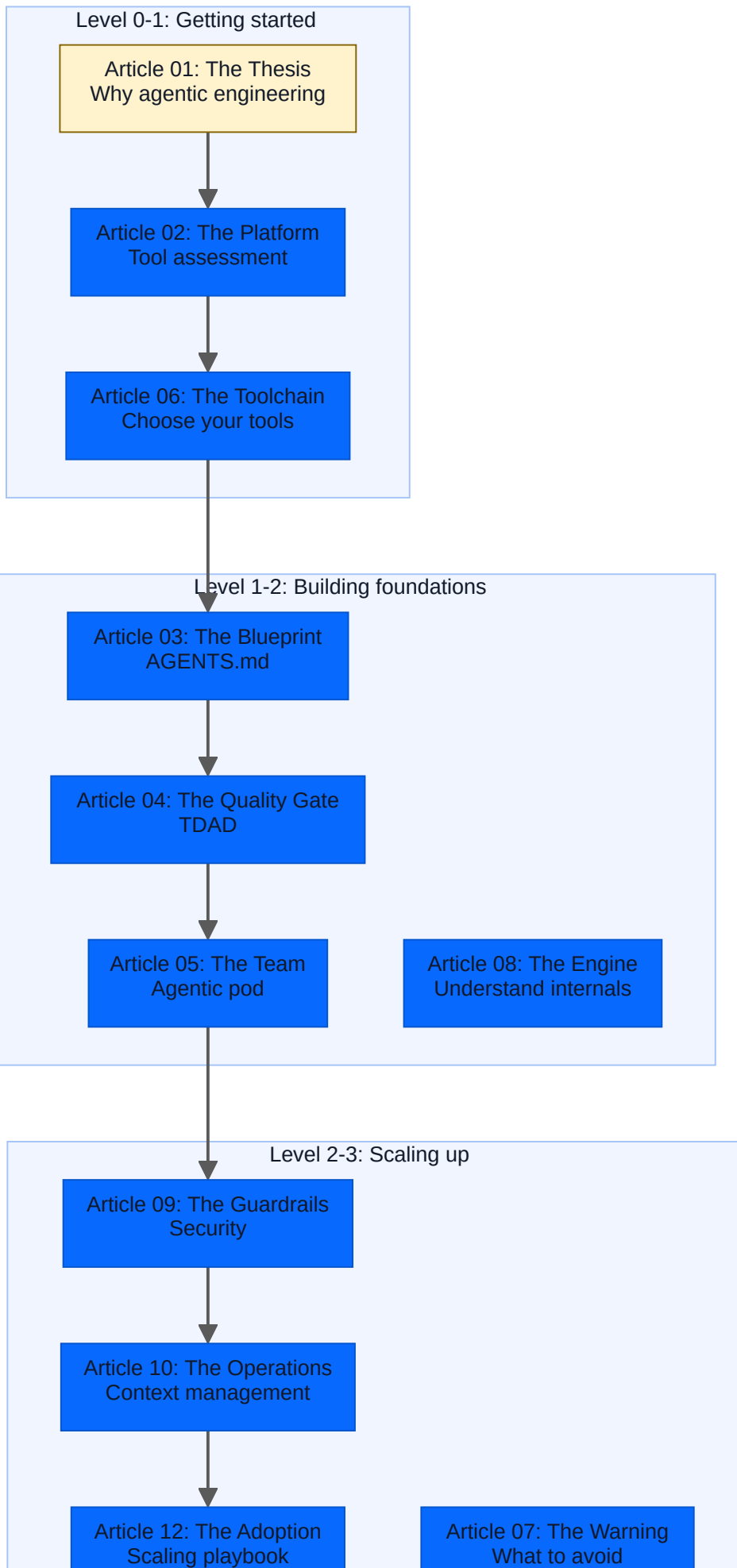
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# Agentic Engineering Maturity Assessment – Q2 2026

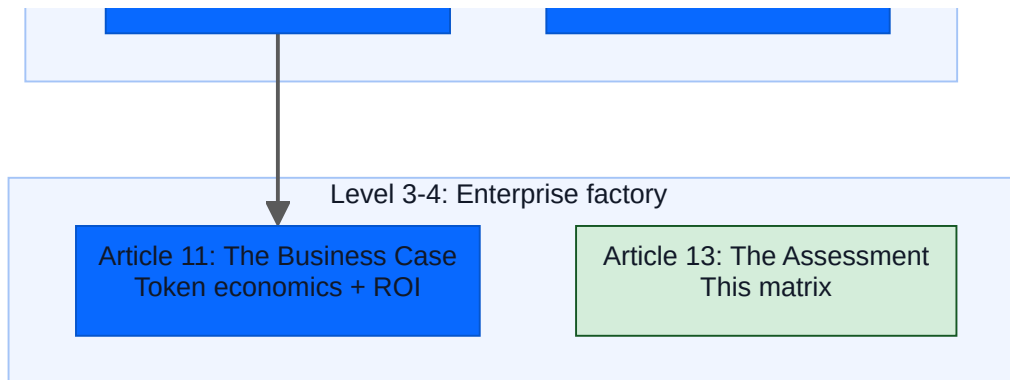
| Dimension          | Q1 Score | Q2 Score | Target (EOY) | Trend | Bottleneck? |
|-----|-----|-----|-----|-----|-----|
| Culture             | 2        | 2        | 3            | →     |              |
| Team Structure      | 1        | 2        | 3            | ↑     |              |
| Agent Instructions  | 1        | 2        | 3            | ↑     |              |
| Quality & Testing   | 1        | 1        | 3            | →     | Δ           |
| Security            | 2        | 2        | 3            | →     |              |
| Toolchain          | 2        | 2        | 3            | →     |              |
| Cost Governance     | 0        | 1        | 3            | ↑     | Δ           |
| Context Management  | 0        | 1        | 2            | ↑     |              |
| Delivery            | 2        | 2        | 3            | →     |              |
| Measurement         | 0        | 1        | 3            | ↑     | Δ           |

**Primary constraint**: Quality & Testing (Level 1)
**Secondary constraints**: Cost Governance (Level 1), Measurement (Level 1)
**Q3 focus**: Advance Quality to Level 2 (TDAD adoption), deploy cost telemetry hooks
```

The matrix as a map of the series

Every cell in the matrix points back to the series. The matrix tells you where to focus. The surrounding articles tell you how to move.





The shift

The matrix gives you a better sentence than 'we are using AI tools'. You can say, for example, that Delivery is Level 2, Quality is Level 1 and the testing gap is the current bottleneck. That is a management statement. It names the problem and implies the next action.

It also forces honesty. If `AGENTS.md` exists but nobody keeps it current, you are not further along because the file exists. If cost telemetry is partial and nobody trusts it, you have not solved cost governance. If leadership says the programme is working but nobody can show the delta in cycle time and defect rate, you do not have measurement.

That is why the constraint matters more than the average. A flat Level 2 profile is healthier than a flashy profile that mixes Level 4 ambition with Level 1 discipline. The weak point always sets the pace.

This closes *From Experiment to Factory*. The series has argued that agentic engineering is not a tool rollout but an organisational system. This final article turns that claim into an assessment instrument. Use it to find the weakest dimension. Use the linked article to improve it. Run the review again next quarter. The factory is not finished when the matrix appears balanced. It is finished only when the organisation stops learning, and that is the point at which it starts to decay.

The Agentic Engineering Series

From experiment to enterprise, building the factory for AI-assisted software engineering at scale.

	Article	Role
1	Codex CLI at One Year	The Platform
2	Agentic Engineering Is Not Vibe Coding	The Wake-Up Call
3	The Agentic Pod	The Team Model
4	TDAD and the Testing Revolution	The Quality Gate
5	The AGENTS.md Playbook	The Blueprint
6	Inside the Machine	The Engine
7	Complete Guide to Codex Security	The Guardrails
8	Context Compaction and Memory	The Efficiency Layer
9	Three Terminals, Three Fates	The Toolchain
10	AI Slopededdon	The Risk
11	Token Economics and ROI	The Business Case
12	The Scaling Playbook	The Rollout
13	The Agentic Engineering Maturity Matrix	The Assessment

1. Container Solutions, "Cloud Native Maturity Matrix." Ten dimensions, Culture, Product, Delivery, Process, Team, Architecture, Reliability, Provisioning, Infrastructure and Security, across five maturity levels. <https://info.container-solutions.com/cloud-maturity-matrix> (<https://info.container-solutions.com/cloud-maturity-matrix>)  ²
2. Goldratt, E.M., *The Goal* (1984). The Theory of Constraints holds that any system's throughput is limited by its tightest constraint. Improving non-constraints does not improve the system until that constraint moves. 
3. METR, "Measuring the Impact of AI Tools on Developer Productivity", randomised controlled trial, February to June 2025. Developers predicted a 24 per cent speed-up, experienced a 19 per cent slowdown and still reported feeling 20 per cent faster. A 39-percentage-point perception gap. Cross-referenced in Article 11. <https://metr.org/blog/2026-02-24-uplift-update/> (<https://metr.org/blog/2026-02-24-uplift-update/>) 
4. Palmer, A., "Amazon AI outage costs estimated 6.3 million lost orders", CNBC, March 2026. AI-assisted code reached production without senior review. Cross-referenced in Article 02. 
5. Bratton, L., "Uber CTO Shows How Claude Code Can Blow Up AI Budgets", *The Information*, April 2026. Uber exhausted its annual AI budget by April. Cross-referenced in Article 11. <https://www.theinformation.com/newsletters/applied-ai/uber-cto-shows-claude-code-can-blow-ai-budgets> (<https://www.theinformation.com/newsletters/applied-ai/uber-cto-shows-claude-code-can-blow-ai-budgets>) 